DG INTERNS HUB

Cyber Security Internship Program

WEEK 1 REPORT

Lab Setup, Linux and Networking Foundations

Name	Harini Edward
Intern ID	DG/15/OCT/CYBER/040
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Topics	Virtual lab, Linux command line, networking basics, TryHackMe, OverTheWire Bandit

Submitted by Harini Edward

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1. Introduction

This report covers everything I did in Week 1 of the DG Interns Hub Cyber Security Internship. The goal of the week was to build a safe home lab, get comfortable with the Linux command line, and understand the networking foundations that all later security work depends on.

I worked through the theory first, then moved to the hands-on tasks: building the lab, practising Linux, mapping the network, and solving OverTheWire Bandit levels. I also completed the assigned TryHackMe rooms, made a Linux cheat sheet and wrote two LinkedIn posts. Each task has its own section in this report, with the screenshots that prove the work.

Objectives of the week

- Build a safe, isolated home lab with virtual machines.
- Learn the Linux command line, file permissions and basic administration.
- Understand IP, DNS, ports, the OSI and TCP/IP models and how a network is laid out.
- Learn the CIA triad, how SOC and VAPT roles differ, and the legal side of security work.
- Practise on TryHackMe and OverTheWire Bandit.

A note on privacy

Some screenshots contained personal details, such as my Windows username, my computer name, hardware addresses, my IPv6 address, one public internet address and my profile picture on TryHackMe, and the Bandit screenshots contained level passwords. I covered or replaced all of these before adding the screenshots to this report. The private addresses that remain (192.168.x.x and 10.x.x.x) only work inside my lab or my local network.

2. Summary of Tasks

The table lists every task from the Week 1 brief, what I did for it, and the section of this report that explains it.

Task	What I did	Status	Section
Theory notes	Answered all 8 theory questions, one note for each	Done	3
Task 1: Build the lab	Kali Linux and Ubuntu Server VMs on a Host-only network, ping tests, a snapshot of each clean VM. My Windows laptop was the third machine	Done, with one change (see section 4)	4
Task 2: Linux practice	3 users, 2 groups, shared folder, find, grep, ps, ss, cron job, 30 commands with output	Done	5
Task 3: Network map	IP, gateway and DNS of each machine, network diagram in draw.io	Done	6
Task 4: Bandit writeups	Levels 0 to 10, one writeup for each level	Done	7
TryHackMe rooms	8 rooms completed to 100 percent, with writeups	Done	8
Task 5: Cheat sheet	Linux Command, What It Does, When to Use It table with 44 entries	Done	9
LinkedIn	Concept post and weekly post	Done	10
Submission	Week 1 folder shared through Google Drive	Done	13

3. Theory Notes

The theory part had eight questions. I wrote each answer as a separate note and saved them in the Notes folder. This section is a summary of those notes, organised question by question.

3.1 Cybersecurity and the CIA triad

Cybersecurity means protecting computers, networks, applications and data from attacks, damage and unauthorized access. It is not only about stopping hackers. It also covers mistakes, such as deleting a database by accident, and failures, such as a server going down. In simple words, it means making sure the right people can use the right information at the right time, and nobody else can.

The CIA triad is the base model of security. Every control we use is trying to protect at least one of these three things.

Part	Meaning	Real-world example	Common control
Confidentiality	Only authorized people can see the data	Chats on WhatsApp are end-to-end encrypted, so even the company cannot read them	Encryption, access control
Integrity	Data stays accurate and is not changed without permission	A UPI payment of Rs. 5,000 must not become Rs. 50,000 on the way	Hashes, digital signatures
Availability	Systems and data can be used when needed	A ticket booking site that stays online when bookings open	Backups, redundancy

The three parts sometimes pull against each other. Very strict access control helps confidentiality but can hurt availability. Different systems also care about different parts: a bank cares most about integrity, while a news website cares more about availability.

3.2 SOC analyst, penetration tester and incident responder

All three roles protect an organization, but they look at security from different sides. The SOC analyst watches, the penetration tester attacks with permission, and the incident responder cleans up when something has gone wrong.

	SOC analyst	VAPT / penetration tester	Incident responder
Goal	Detect and triage threats early	Find weaknesses before real attackers do	Contain, investigate and recover from an incident
Daily work	Monitor alerts, check logs, decide if an alert is real, escalate	Scan, test manually, exploit in a controlled way, write a report with fixes	Isolate machines, collect evidence, find the root cause, remove the threat
Tools	SIEM (Splunk, Wazuh, QRadar), EDR, Wireshark	Nmap, Burp Suite, Metasploit, Kali Linux	Volatility, Autopsy, FTK Imager, EDR, SIEM
Team	Blue team	Red team	Blue team

SOC Level 1 is the usual entry point for freshers, and the job is mostly triage. A penetration tester always needs written permission and a defined scope. Incident responders work close to digital forensics, because evidence handling matters in legal cases.

3.3 OSI model versus TCP/IP model

Both models explain how data travels from one device to another. OSI is the theoretical model with 7 layers, and TCP/IP is the practical model the internet runs on, with 4 layers.

OSI layer	What it does	Example protocols	One attack
7 Application	Interface for user programs	HTTP, HTTPS, FTP, SMTP, DNS	SQL injection, phishing
6 Presentation	Encryption, encoding, compression	SSL/TLS	SSL stripping
5 Session	Opens, manages and closes sessions	NetBIOS, RPC	Session hijacking
4 Transport	End-to-end delivery, ports	TCP, UDP	SYN flood
3 Network	Logical addressing and routing	IP, ICMP, IPsec	IP spoofing
2 Data Link	MAC addressing, frames, switching	Ethernet, ARP	ARP spoofing
1 Physical	Raw bits over cable or wireless	Cables, Wi-Fi	Wiretapping

Point	OSI	TCP/IP
Number of layers	7	4 (Application, Transport, Internet, Network Access)
Type	Theoretical reference model	Practical model used on the internet
Session and presentation	Separate layers	Merged into the application layer

3.4 IP, MAC, DNS, DHCP and ports

Term	What it is	Example
IP address	Logical address of a device on a network. It can change	192.168.56.101
MAC address	Physical hardware address of the network card, fixed by the manufacturer	Six pairs of hex digits
DNS	Turns a name into an IP address	google.com becomes an IP address
DHCP	Gives a device its IP, gateway and DNS automatically (Discover, Offer, Request, Acknowledge)	A phone joining Wi-Fi
Port	A number that identifies a service on a device, from 0 to 65535	443 for HTTPS

I also made a table of 12 common ports and the risk that goes with each one:

Port	Service	Typical risk
21	FTP	Plain text login, anonymous access left on
22	SSH	Brute force attacks, weak passwords
23	Telnet	No encryption, easy to sniff
25	SMTP	Spam relay, email spoofing, phishing
53	DNS	Poisoning, tunneling, amplification attacks
80	HTTP	Unencrypted traffic, web attacks
110	POP3	Plain text mail login
143	IMAP	Plain text mail login
443	HTTPS	Weak TLS settings, web application flaws
445	SMB	Exposed shares, EternalBlue and WannaCry
3306	MySQL	Exposed database, weak root password
3389	RDP	Brute force, ransomware entry point

3.5 Linux file system and file permissions

Linux has one tree that starts at / (root), and there are no drive letters like C: or D:.

Directory	What it holds	Why it matters in security
`/etc`	System and program configuration files	Attackers target it for passwords and settings
`/var/log`	Log files such as auth.log and syslog	First place a SOC analyst checks for logins and errors
`/home`	Personal folders of users	User files, SSH keys, shell history
`/tmp`	Temporary files that anyone can write to	Malware often drops files here

Running `ls -l` shows permissions like `rwxr-xr--`. The three groups are owner, group and others, and each has read (4), write (2) and execute (1). So `rwxr-xr--` is 754.

```
chmod 755 script.sh # owner rwx, group r-x, others r-x
chmod u+x script.sh # add execute for the owner
sudo chown ubuntu:devs file # change owner and group
chmod u+s program # set the SUID bit
find / -perm -4000 2>/dev/null # list SUID files
```

The SUID bit makes a program run with the rights of its owner instead of the person running it. This is how a normal user can change their own password. If a SUID program owned by root has a flaw, an attacker can use it to become root, so checking SUID files is a standard step in privilege escalation and hardening.

3.6 Threat, vulnerability, risk and exploit

Term	Meaning	Example
Threat	Anything that could cause harm	A ransomware group targeting hospitals
Vulnerability	A weakness that a threat can use	An unpatched Windows server with an SMB flaw
Exploit	The method or code that uses the weakness	The EternalBlue exploit
Risk	The chance that a threat uses a weakness, combined with the damage it causes	An internet-facing hospital server with that flaw has a high chance of ransomware and data loss

A simple way to remember it is a house: the burglar is the threat, the broken back-door lock is the vulnerability, the crowbar is the exploit, and the risk is how likely the break-in is and how much you would lose.

3.7 VM networking modes

Mode	Internet	VM to VM	Best use
NAT	Yes	No	Updates and downloads for one VM
NAT Network	Yes	Yes	A lab where VMs talk to each other and also get updates
Host-only	No	Yes	A lab that is cut off from the internet and from the real network
Bridged	Yes	Yes	A VM that must act like a real device on the network

A vulnerable lab must never be bridged. In bridged mode the vulnerable machine sits on the real network, so anyone on the same Wi-Fi can reach it, and if it is compromised the attacker can move to real laptops, phones and the router. Tools such as Kali could also scan other people's devices by mistake. For this reason I used a Host-only network for the lab, with a separate NAT adapter only for updates.

3.8 Legal and ethical basics

Term	Meaning
Authorization	Written permission from the owner of the system. Without it, testing is illegal
Scope	The exact systems, IPs, apps and times that may be tested. Anything outside is off limits
Rules of Engagement	The agreed rules for the test: allowed techniques, hours, emergency contact, data handling

Law in India	Relevance
IT Act 2000, Section 43	Unauthorized access, data damage, introducing malware
IT Act 2000, Section 66	Dishonest or fraudulent acts under Section 43 are criminal offences
IT Act 2000, Sections 66C, 66D, 66F, 72	Identity theft, cheating by impersonation, cyber terrorism, breach of confidentiality
Digital Personal Data Protection Act 2023	Rules for handling personal data, with penalties for breaches
Bharatiya Sakshya Adhiniyam 2023, Section 63	Electronic records as evidence, with the required certificate
CERT-In directions 2022	Reporting of cyber incidents within 6 hours and keeping logs for 180 days

The main lesson from this topic is that good intention is not a legal defence. Authorization is. Even a simple port scan on a system I do not own can be treated as unauthorized access.

4. Task 1: Build the Lab

What the task asked

Install VirtualBox, create a Kali Linux VM, an Ubuntu VM and a Windows 10 or 11 evaluation VM, put them on one Host-only or NAT Network, ping between them and take a snapshot of each clean VM. Screenshots were needed for the VM list, each VM's IP and the successful pings.

What I did

I created two virtual machines in VirtualBox. In the VM list they are named Kali-CLI and Ubuntu-lab, and inside the machines their hostnames are kali-lab and ubuntu-lab. Both are on a Host-only network, 192.168.56.0/24, and both also have a NAT adapter for internet access. I did not create a Windows VM because my laptop did not have enough storage. My own Windows laptop is the third machine in the lab, and its Host-only adapter has the address 192.168.56.1. I can also log in to the Ubuntu VM from Windows with SSH.

I checked the IP address on each VM and pinged between the machines. When the clean installation was ready, I took a snapshot of each VM, named clean install.

Machine	Host-only IP	Role in the lab
Windows laptop	192.168.56.1	Third machine, SSH client
Kali VM (kali-lab)	192.168.56.101	Security tools machine
Ubuntu VM (ubuntu-lab)	192.168.56.102	Linux practice server

Because the NAT adapter gives internet access, this lab is isolated from my real network but it is not offline. For vulnerable machines in later weeks, the NAT adapter should be turned off.

Evidence

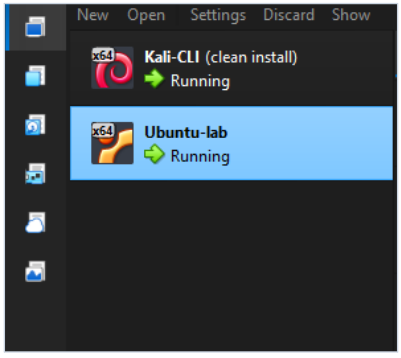


Figure 1: VM list in VirtualBox, Kali-CLI and Ubuntu-lab both running

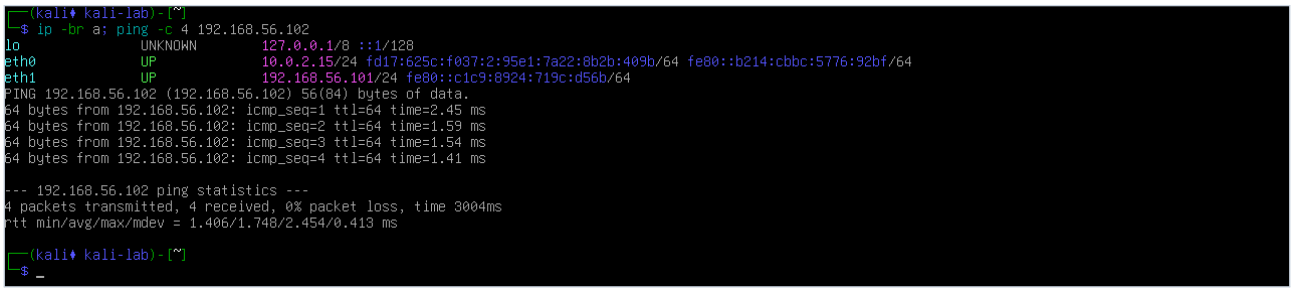


Figure 2: Kali VM IP addresses and a ping to the Ubuntu VM (192.168.56.102), 4 packets sent and 4 received

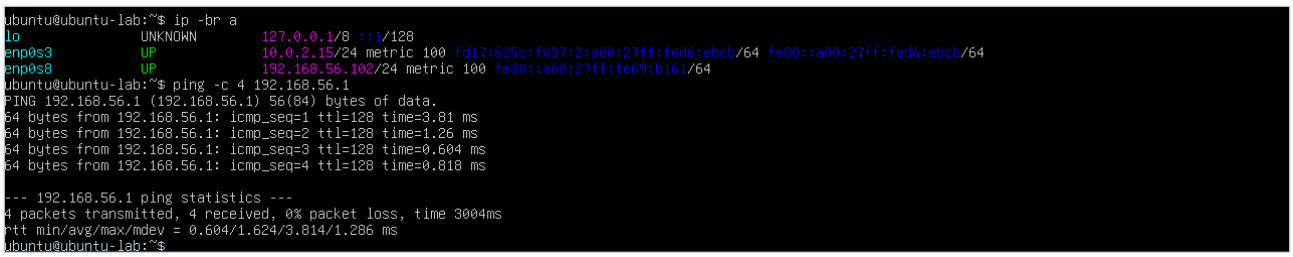


Figure 3: Ubuntu VM IP addresses and a ping to the Windows laptop (192.168.56.1)

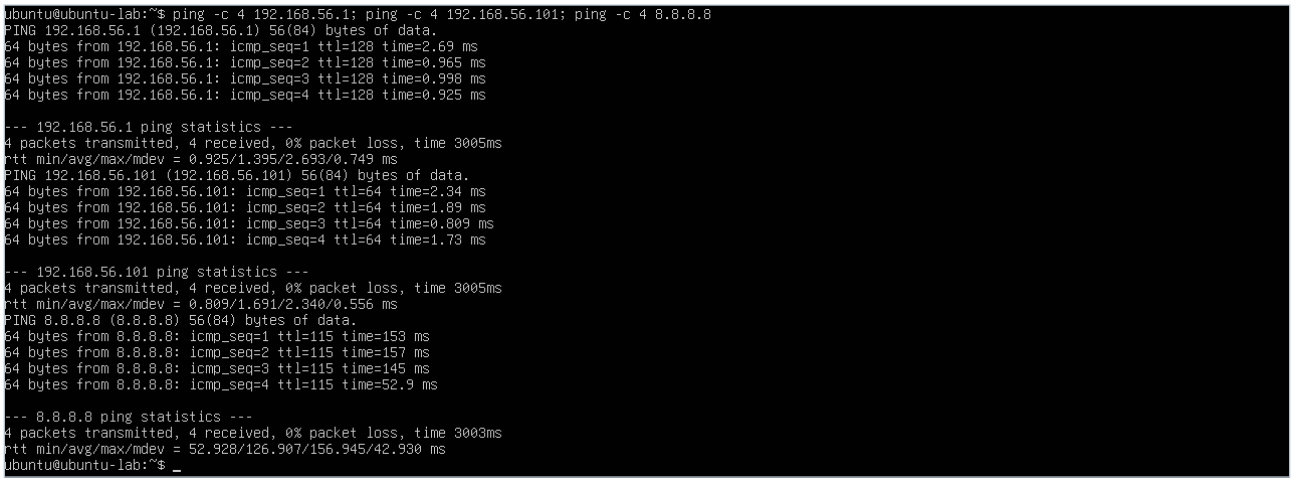


Figure 4: Ubuntu VM pinging the Windows laptop, the Kali VM and the internet (8.8.8.8), all with 0 percent packet loss

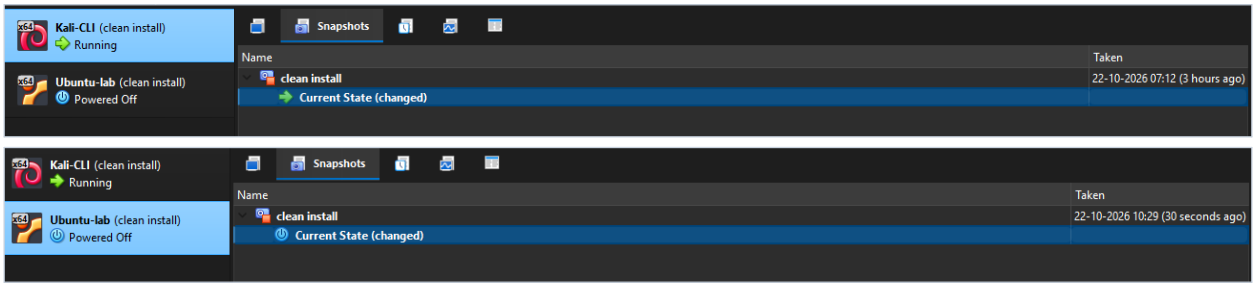


Figure 5: Snapshots named clean install for the Kali VM (top) and the Ubuntu VM (bottom)

Result

The Kali VM, the Ubuntu VM and the Windows laptop can reach each other over the Host-only network, and the Ubuntu VM can also reach the internet. The only difference from the brief is the Windows machine, which is my laptop instead of a VM.

5. Task 2: Linux Practice

What the task asked

On Ubuntu, create 3 users and 2 groups, create a shared folder with correct permissions, use find, grep, ps and ss -tulpn, schedule a cron job, and document 30 commands with their output.

What I did

Requirement	What I did	Evidence
3 users	Created alice, bob and carol with adduser	Figure 6, Table 5.1
2 groups	Created developers (members alice and bob) and analysts (member carol) with groupadd and usermod	Figure 6, Table 5.1
Shared folder	Created /srv/shared, set the owner to root and the group to developers, and set the mode to 2770 (drwxrws---)	Figure 7
Permission test	alice created a file in the folder. carol, who is not in developers, was denied	Figure 7, Table 5.1
find, grep, ps	find /etc -name "*.conf" -type f, grep "ubuntu" /etc/passwd, ps aux	Figure 8
ss -tulpn	Listed the listening ports. sshd listens on port 22 and DNS on port 53	Figure 8
Cron job	Scheduled a job that adds the date to /home/ubuntu/cron-test.log every minute. The log shows 8 lines, 12:35 to 12:42. I removed the job with crontab -r after the test	Figure 9
30 commands with output	Recorded in Table 5.1 below and in a separate file in the Findings folder	Table 5.1

Evidence

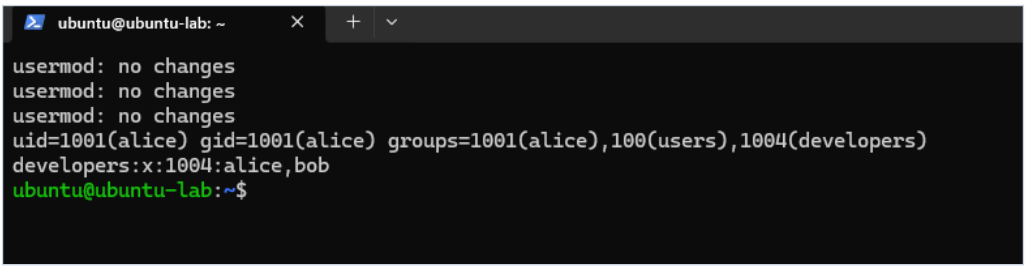


Figure 6: Output of the group commands: usermod (no changes, because the users were already in their groups), id alice and getent group developers. The commands are rows 13 to 17 of Table 5.1

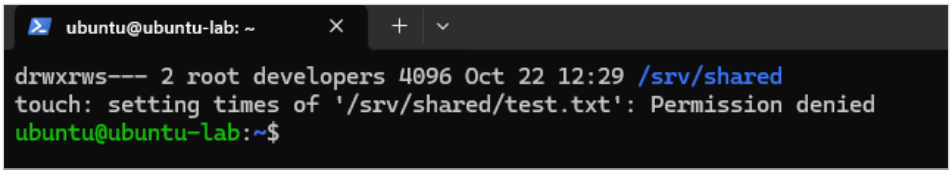


Figure 7: Output of ls -ld /srv/shared (owned by root, group developers) and of carol's attempt to create a file, which was denied. The commands are rows 21 and 23 of Table 5.1

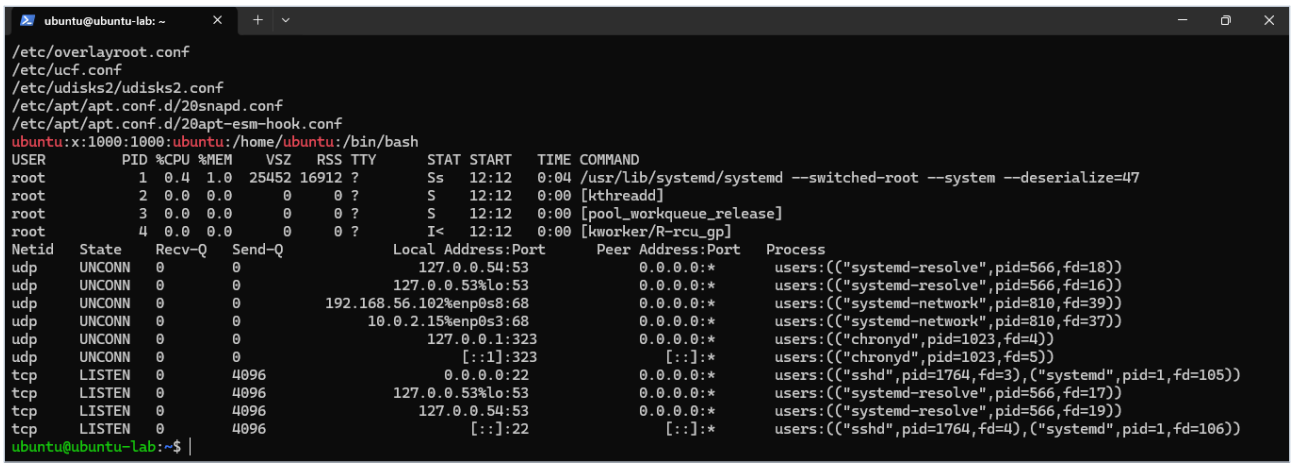


Figure 8: Output of find, grep, ps and ss -tulpn

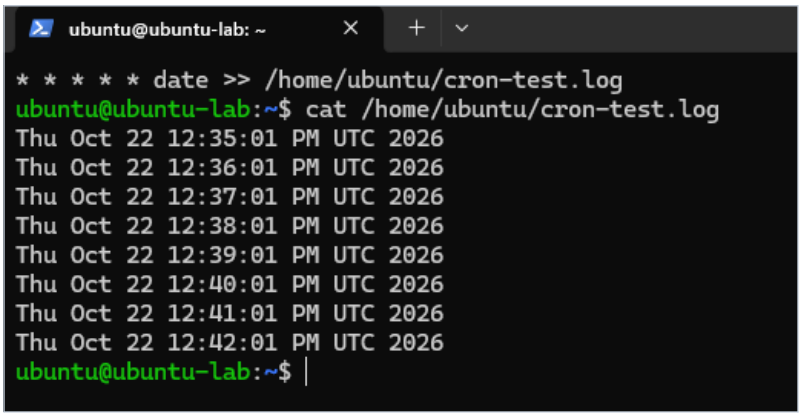


Figure 9: The cron job adds the date to a log file every minute, and the log shows a new line each minute

What I learned from this task

The folder mode 2770 gives the owner and the group full access and gives everyone else none. The test with carol showed this in practice: she is not in developers, so the system refused to create her file. The s in drwxrws--- is the SGID bit, which makes new files inside the folder keep the group developers. The ss output showed that SSH is the main service listening on the Ubuntu VM.

Table 5.1: 30 commands with output (Ubuntu-lab)

#	Command	What it does	Output (short)
1	`whoami`	Shows current user	ubuntu
2	`hostname`	Shows machine name	ubuntu-lab
3	`pwd`	Shows current folder	/home/ubuntu
4	`uname -r`	Shows kernel version	7.0.0-38-generic
5	`uptime`	Shows running time and load	up 12 min, 2 users, load 0.00
6	`df -h`	Shows disk space	/ is 8.1G, 60% used
7	`free -h`	Shows memory	1.6Gi total, 1.2Gi available
8	`sudo adduser --disabled-password --gecos "" alice`	Creates user alice	User created, uid 1001
9	`sudo adduser --disabled-password --gecos "" bob`	Creates user bob	User created
10	`sudo adduser --disabled-password --gecos "" carol`	Creates user carol	User created
11	`sudo groupadd developers`	Creates group	Group created (gid 1004)
12	`sudo groupadd analysts`	Creates group	Group created
13	`sudo usermod -aG developers alice`	Adds alice to group	usermod: no changes (already a member, repeated run)
14	`sudo usermod -aG developers bob`	Adds bob to group	usermod: no changes (already a member, repeated run)
15	`sudo usermod -aG analysts carol`	Adds carol to group	usermod: no changes (already a member, repeated run)
16	`id alice`	Shows user and groups	uid=1001(alice) ... 1004(developers)
17	`getent group developers`	Lists group members	developers:x:1004:alice,bob
18	`sudo mkdir /srv/shared`	Creates shared folder	No output (success)
19	`sudo chown root:developers /srv/shared`	Sets owner and group	No output (success)
20	`sudo chmod 2770 /srv/shared`	Sets permissions with SGID	No output (success)
21	`ls -ld /srv/shared`	Shows folder permissions	drwxrws--- 2 root developers ... /srv/shared
22	`sudo -u alice touch /srv/shared/alice-note.txt`	alice creates a file	Success, no output
23	`sudo -u carol touch /srv/shared/test.txt`	carol tries to create a file	Permission denied (expected)
24	`find /etc -name "*.conf" -type f, then head -5`	Finds .conf files	5 paths, e.g. /etc/ucf.conf
25	`grep "ubuntu" /etc/passwd`	Searches text in a file	ubuntu:x:1000:1000:ubuntu:/home/ubuntu:/bin/bash
26	`ps aux, then head -5`	Lists running processes	systemd (PID 1), kthreadd, and others
27	`sudo ss -tulpn`	Lists listening ports	sshd on port 22, DNS on 53

#	Command	What it does	Output (short)
28	`crontab - (add job)`	Schedules a job every minute	No output (success)
29	`crontab -l`	Shows the cron jobs	* * * * * date >> /home/ubuntu/cron-test.log
30	`cat /home/ubuntu/cron-test.log`	Shows the cron result	8 date lines, 12:35 to 12:42

Extra commands used: `sudo -v` (cached sudo login) and `crontab -r` (removed the job after the test).

6. Task 3: Network Map

What the task asked

Record the IP, gateway and DNS of each VM using ip a, ping, traceroute and nslookup or dig, and then draw the lab network diagram in draw.io.

What I did

On Kali and Ubuntu I used ip and ip route to find the addresses and the default gateway, and I checked the DNS servers in /etc/resolv.conf and with resolvectl. I used ping and traceroute to test connectivity and the path to the internet, and nslookup and dig to check name lookups. On the Windows laptop I used ipconfig, ping, tracert and nslookup. Then I drew the map in draw.io.

Machine	Adapter	IP address	Gateway	DNS
Windows laptop	Host-only	192.168.56.1	None	-
Windows laptop	Wi-Fi	10.210.2.193	10.210.2.76	10.210.2.76
Kali VM	eth1, Host-only	192.168.56.101	-	-
Kali VM	eth0, NAT	10.0.2.15	10.0.2.2	10.210.2.76
Ubuntu VM	enp0s8, Host-only	192.168.56.102	-	-
Ubuntu VM	enp0s3, NAT	10.0.2.15	10.0.2.2	10.210.2.76

A dash means nothing is set on that adapter. Both VMs show 10.0.2.15 on the NAT adapter because each VM has its own separate NAT network.

6.1 IP address, gateway and DNS

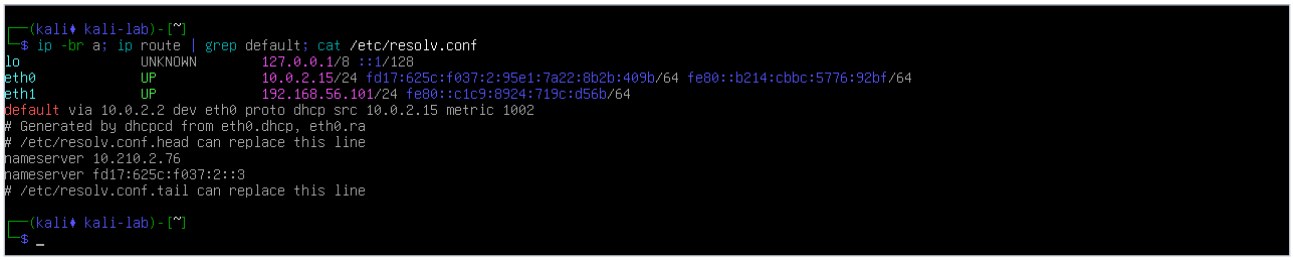


Figure 10: Kali VM. The default gateway is 10.0.2.2 and the DNS server is 10.210.2.76



Figure 11: Ubuntu VM. The default gateway is 10.0.2.2 and the DNS server is 10.210.2.76

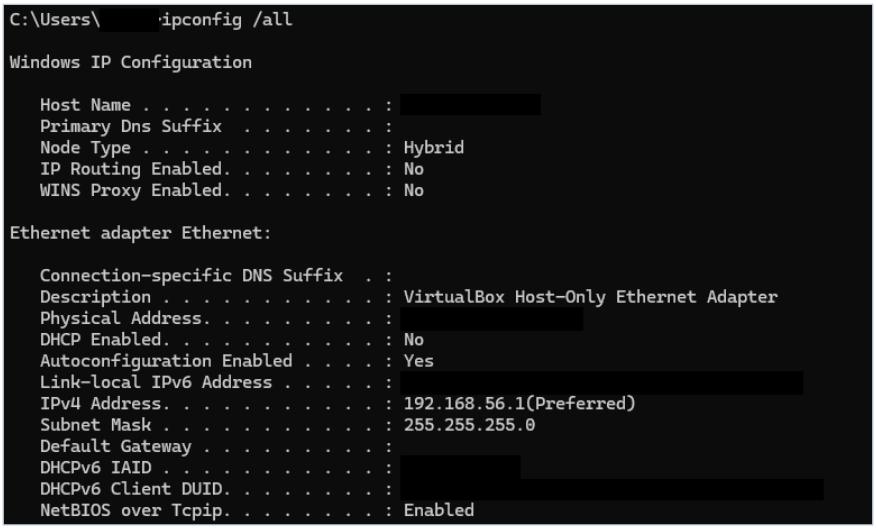


Figure 12: Windows laptop, Host-only adapter 192.168.56.1 with no gateway. My computer name, hardware addresses and IPv6 details are hidden

6.2 Ping and traceroute

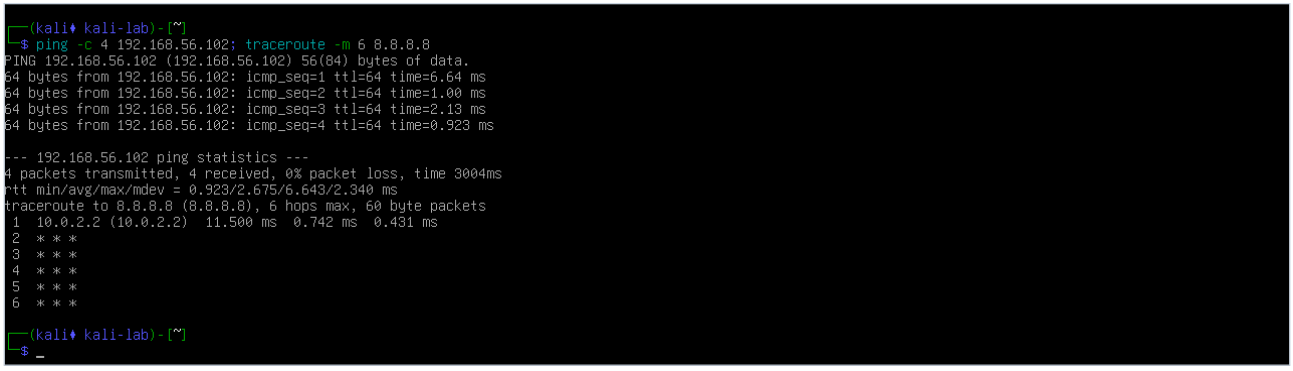


Figure 13: Kali VM. The ping to the Ubuntu VM works. In traceroute the first hop is the NAT gateway 10.0.2.2 and the later hops did not reply

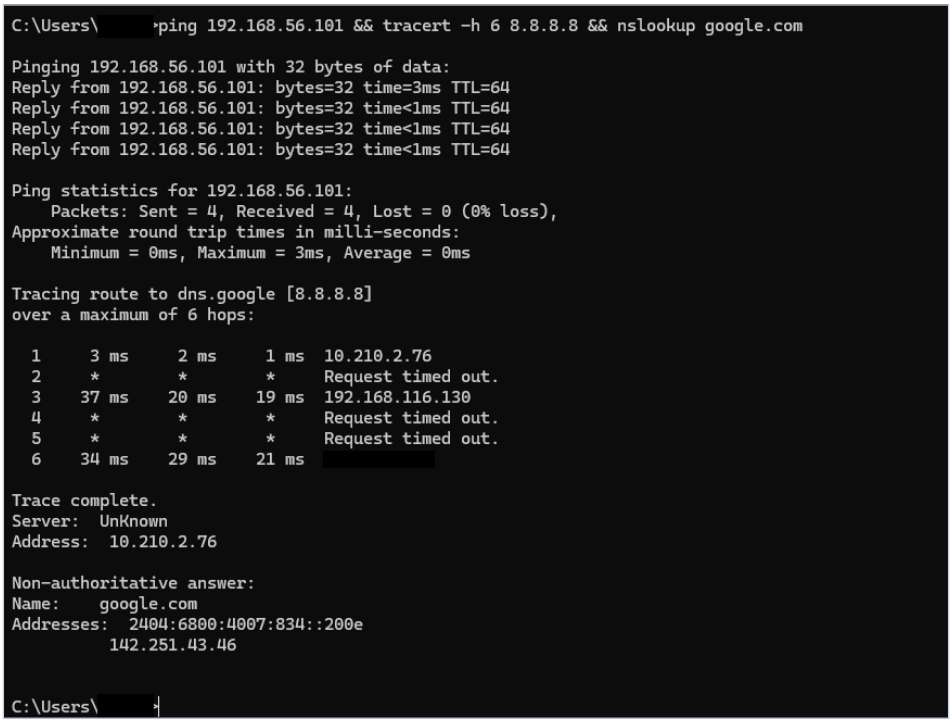


Figure 14: Windows laptop. The ping to the Kali VM works, and tracert goes through my Wi-Fi gateway 10.210.2.76 first. One public address in hop 6 is hidden

6.3 DNS lookups

```
┌──(kali㉿kali-lab)-[~]
└─$ nslookup google.com; dig google.com
Server:         10.210.2.76
Address:        10.210.2.76#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.43.46
Name:   google.com
Address: 2404:6800:4007:834::200e


; <<>> DiG 9.20.29-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12172
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             221     IN      A       142.251.43.46

;; Query time: 20 msec
;; SERVER: 10.210.2.76#53(10.210.2.76) (UDP)
;; WHEN: Thu Oct 22 14:49:18 IST 2026
;; MSG SIZE  rcvd: 44


┌──(kali㉿kali-lab)-[~]
└─$
```

Figure 15: Kali VM looking up google.com with nslookup and dig. The server used is 10.210.2.76

```
ubuntu@ubuntu-lab:~$ traceroute -m 6 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 6 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  0.823 ms  0.694 ms  0.763 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
ubuntu@ubuntu-lab:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.206.14
Name:   google.com
Address: 2404:6800:4009:824::200e

ubuntu@ubuntu-lab:~$ dig google.com

; <<>> DiG 9.20.24-1ubuntu0.4-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22289
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             295     IN      A       142.250.206.174

;; Query time: 86 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Oct 22 14:02:56 UTC 2026
;; MSG SIZE  rcvd: 55

ubuntu@ubuntu-lab:~$ ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             10.0.2.15/24 metric 100 fd17:625c:f037:2:a00:27ff:fed6:ebcb/64 fe80::a00:27ff:fed6:ebcb/64
enp0s8           UP             192.168.56.102/24 metric 100 fe80::a00:27ff:fe69:b161/64
ubuntu@ubuntu-lab:~$ ip route | grep default
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
ubuntu@ubuntu-lab:~$ resolvectl status | grep -i "dns server"
Current DNS Server: 10.210.2.76
       DNS Servers: 10.210.2.76 fd17:625c:f037:2::3
ubuntu@ubuntu-lab:~$
```

Figure 16: Ubuntu VM with traceroute, nslookup and dig. Ubuntu uses its local resolver 127.0.0.53, and the DNS server behind it is 10.210.2.76

6.4 The network diagram

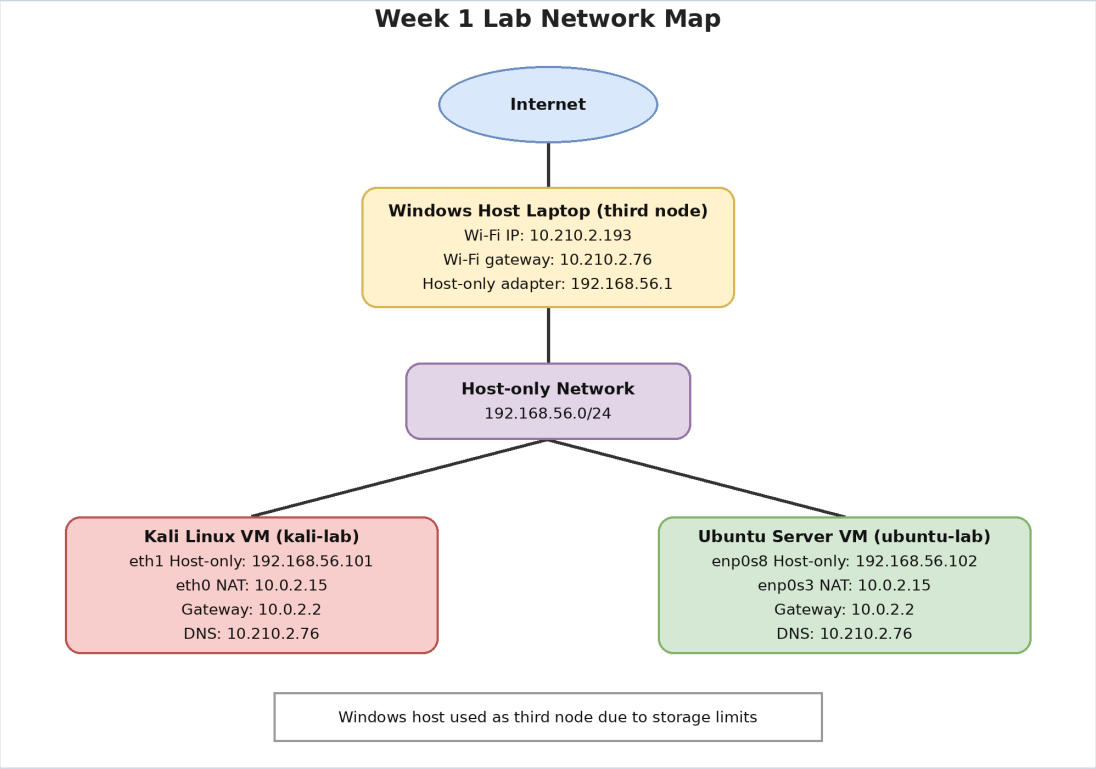


Figure 17: Lab network map drawn in draw.io

What the results show

- On Kali and Ubuntu the default route goes through 10.0.2.2, on the NAT adapter, and the DNS server is 10.210.2.76.
- The Windows Host-only adapter, 192.168.56.1, has no default gateway.
- traceroute to 8.8.8.8 from Kali and Ubuntu shows 10.0.2.2 as the first hop, and the other hops (up to 6) did not reply. From the Windows laptop, tracert shows 10.210.2.76 as the first hop.
- On Ubuntu, nslookup and dig use 127.0.0.53, which is the local resolver, and resolvectl lists 10.210.2.76 as the DNS server.

7. Task 4: OverTheWire Bandit, Levels 0 to 10

What the task asked

Complete Bandit levels 0 to 10 and write a writeup for each level, without publishing any level passwords.

What I did

Bandit is a wargame that teaches the Linux command line. You connect to the game server over SSH, and every level hides the password for the next level in a file or in the output of a command. I solved Levels 0 to 10, wrote a separate writeup for every level, and kept my screenshots as proof.

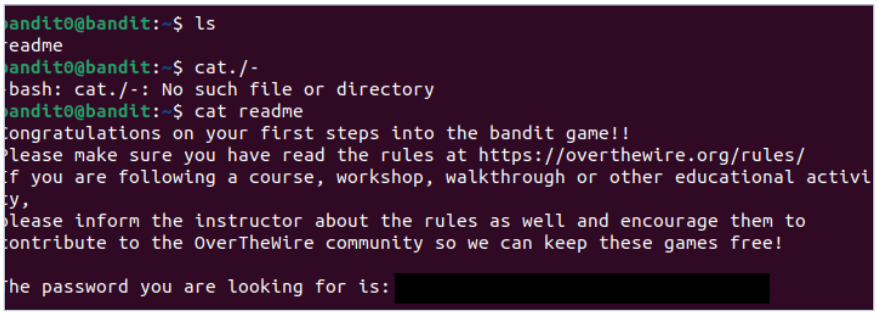
Level	Main command	Technique
0 to 1	`ls, cat readme`	Reading a file
1 to 2	`cat ./-`	A file named with a dash
2 to 3	`cat -- "--spaces in this filename--"`	Spaces and dashes in a name
3 to 4	`ls -la, cat ...Hiding-From-You`	Hidden files
4 to 5	`file ./*, cat ./-file07`	Finding the readable file
5 to 6	`find . -type f -size 1033c ! -executable`	find by size and permission
6 to 7	`find / -type f -user bandit7 -group bandit6 -size 33c 2>/dev/null`	find by owner, group and size
7 to 8	`grep millionth data.txt`	Searching text
8 to 9	`sort data.txt | uniq -u`	The one unique line
9 to 10	`strings data.txt | grep '='`	Readable text in a binary file
10 to 11	`base64 -d data.txt`	Decoding Base64

Level by level

Level 0 to 1

```
ls, cat readme
```

The home folder has a file called readme, and reading it with cat gives the password.



Password hidden

Level 1 to 2

```
cat ./-
```

The file is named with a single dash. Plain cat - would wait for keyboard input, so I opened it with the ./ prefix.

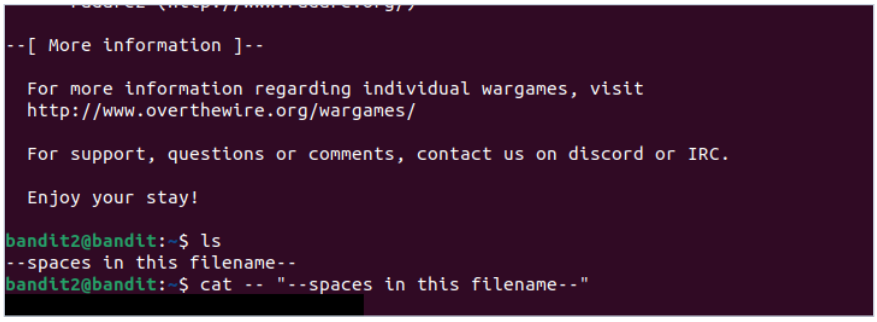


Password hidden

Level 2 to 3

```
cat -- "--spaces in this filename--"
```

The filename has spaces and starts with dashes. Quotes keep it as one name, and -- tells cat that no more options follow.

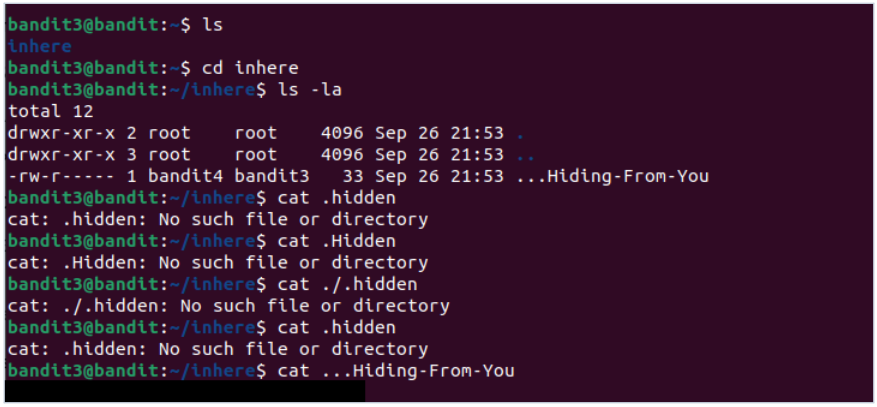


Password hidden

Level 3 to 4

```
ls -la, cat ...Hiding-From-You
```

A normal ls does not show hidden files. ls -la inside the inhere folder showed the hidden file ...Hiding-From-You, which I read with cat. Before that, I tried a few wrong names such as .hidden.

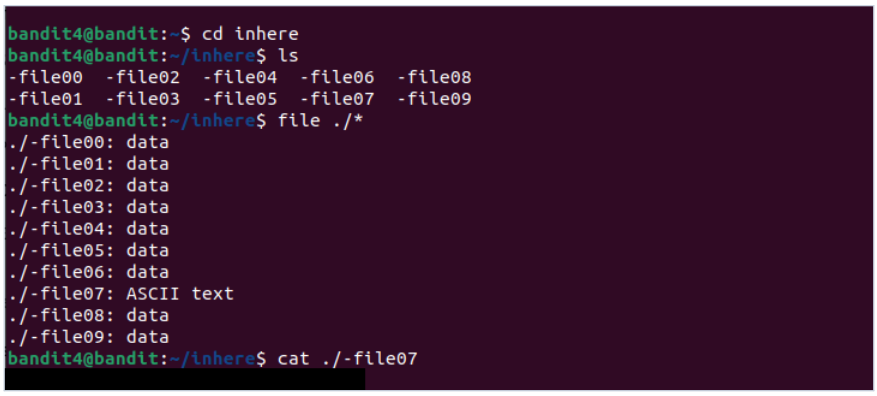


Password hidden

Level 4 to 5

```
file ./*, cat ./-file07
```

All the files start with a dash. Running file on them showed that only -file07 is ASCII text, so that was the readable one.

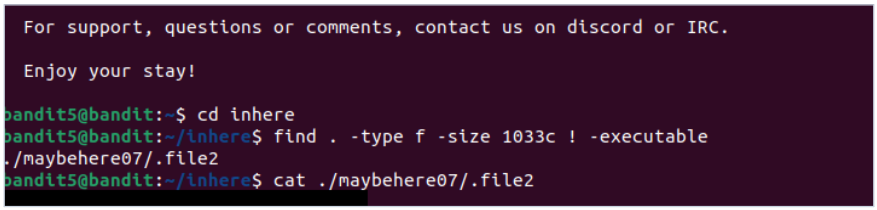


Password hidden

Level 5 to 6

```
find . -type f -size 1033c ! -executable
```

find looked for a file of exactly 1033 bytes that is not executable. It returned ./maybehere07/.file2, which I opened with cat.

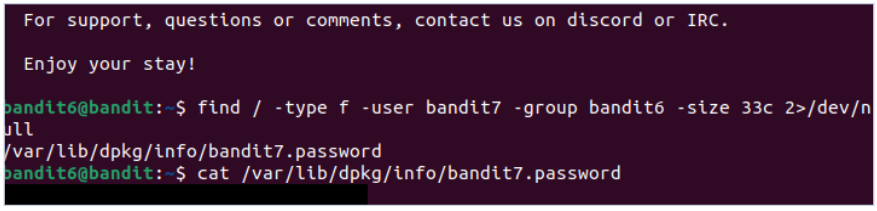


Password hidden

Level 6 to 7

```
find / -type f -user bandit7 -group bandit6 -size 33c 2>/dev/null
```

The search covers the whole system for a file owned by user bandit7 and group bandit6 with a size of 33 bytes. 2>/dev/null hides the permission errors. The file found was /var/lib/dpkg/info/bandit7.password.

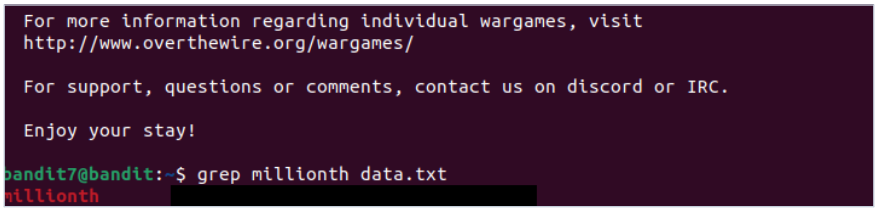


Password hidden

Level 7 to 8

```
grep millionth data.txt
```

grep prints only the line with the word millionth. The password is written next to that word.

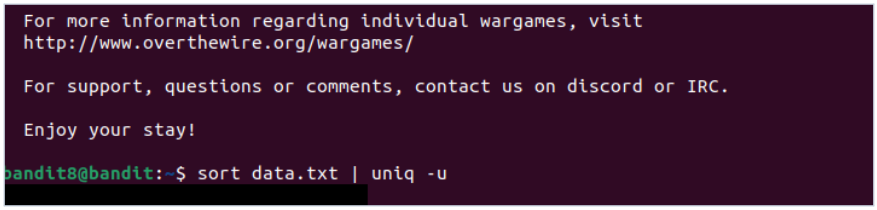


Password hidden

Level 8 to 9

```
sort data.txt | uniq -u
```

sort puts identical lines next to each other, and uniq -u prints only the line that appears once.



Password hidden

Level 9 to 10

```
strings data.txt | grep '='
```

data.txt is a binary file. strings pulls out the readable text, and grep keeps the lines with = signs. The password comes after a long line of equal signs.

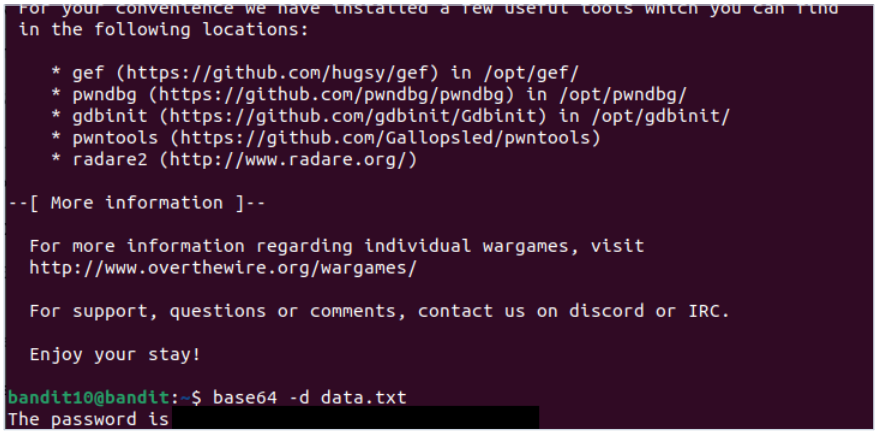


Password hidden

Level 10 to 11

```
base64 -d data.txt
```

The file holds Base64 text. Decoding it with base64 -d prints a sentence that contains the password.



Password hidden

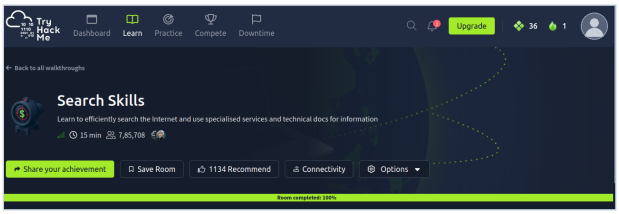
What I learned from Bandit

Bandit taught me to read error messages carefully instead of repeating the same command. It also taught me small habits that matter: put ./ before odd filenames, use quotes for names with spaces, use -- to end options, and use ls -la for hidden files.

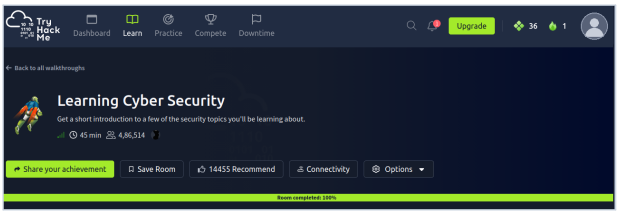
8. TryHackMe Rooms

I completed all 8 assigned rooms to 100 percent and wrote a separate writeup for each one. The writeups are in the Writeups folder, and the completion screenshots are below.

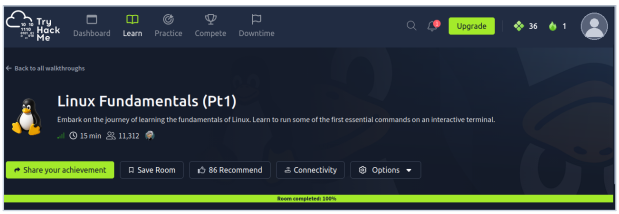
Room	Time	Room focus
Search Skills	15 min	Searching the internet efficiently and using specialised services and technical documents
Learning Cyber Security	45 min	A short introduction to the security topics in the learning path
Linux Fundamentals Part 1	15 min	The first essential Linux commands on an interactive terminal
What is Networking?	30 min	The fundamentals of computer networking
Introductory Networking	20 min	Networking theory and basic networking tools
Networking Concepts	60 min	The ISO OSI model and the TCP/IP protocol suite
Pyramid of Pain	30 min	How indicators are ranked by how painful they are for an attacker to change
Intro to LAN	15 min	The technologies and designs that power private networks



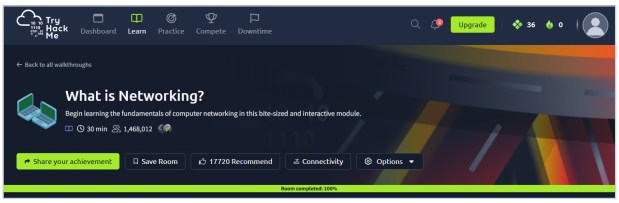
Search Skills



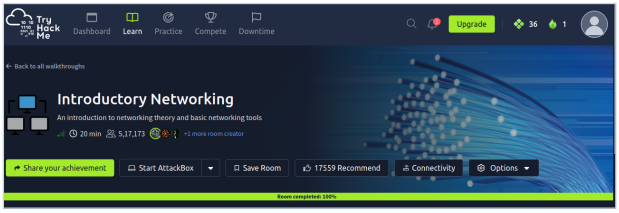
Learning Cyber Security



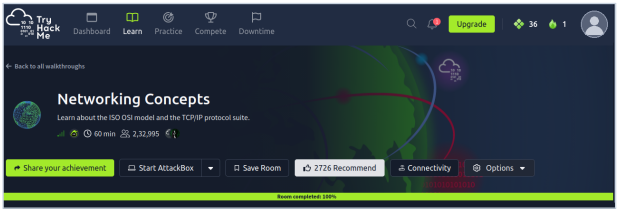
Linux Fundamentals Part 1



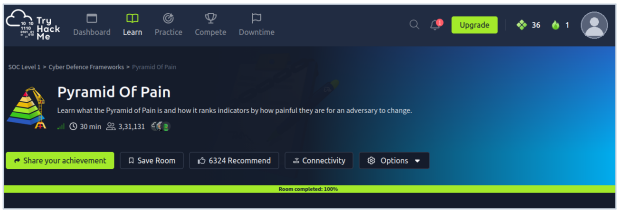
What is Networking?



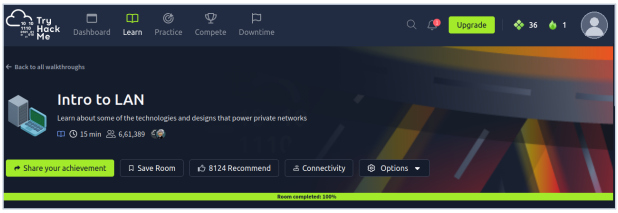
Introductory Networking



Networking Concepts



Pyramid of Pain



Intro to LAN

9. Task 5: Linux Cheat Sheet

What the task asked

Create a table called Linux Command, What It Does, When to Use It with at least 30 entries.

What I did

I made the cheat sheet with three columns: the command, what it does and when to use it. It has 44 entries, which covers file handling, text searching, permissions, users, processes, disk use, networking and package management. The full table is below, and the file is saved in the Cheat Sheet folder.

Linux Command	What It Does	When to Use It
`pwd`	Shows the folder you are in	When you are lost and need to know your location
`ls`	Lists files and folders	To see what is inside a folder
`ls -la`	Lists everything with details, including hidden files	To check permissions, owners and hidden files
`cd`	Changes folder	To move to another directory
`mkdir`	Creates a new folder	When setting up a project or lab folder
`rmdir`	Removes an empty folder	To clean up folders you no longer need
`touch`	Creates an empty file	To quickly make a new file
`cp`	Copies files or folders	To make a backup before editing something
`mv`	Moves or renames files	To rename a file or put it in a new place
`rm`	Deletes files	To remove files you no longer need, carefully, there is no recycle bin
`cat`	Prints a file on screen	To read short files
`less`	Reads a long file page by page	To read logs without flooding the screen
`head`	Shows the first lines of a file	To peek at the start of a file or log
`tail`	Shows the last lines of a file	To see the newest log entries
`tail -f`	Follows a file as it grows	To watch a log live while something runs
`grep`	Searches for text inside files	To find a keyword or error in a file
`find`	Searches for files by name, size or type	When you know a file exists but not where
`wc -l`	Counts lines in a file	To count log entries or results
`sort`	Sorts lines of text	To put output in order
`echo`	Prints text or writes it to a file	To test commands or add a line to a file
`nano`	Simple text editor in the terminal	To edit files without leaving the terminal
`chmod`	Changes file permissions	To make a script runnable or lock a file down
`chown`	Changes file owner	When the wrong user owns a file
`sudo`	Runs a command as administrator	When a command needs admin rights
`whoami`	Shows your current user	To confirm which account you are using
`id`	Shows your user ID and groups	To check what groups and rights you have
`man`	Opens the manual for a command	When you forget how a command works

Linux Command	What It Does	When to Use It
`history`	Lists commands you typed before	To find and repeat an earlier command
`uname -a`	Shows system and kernel information	To record which system you are on
`ps aux`	Lists running processes	To see what is running on the machine
`top`	Shows live CPU and memory use	To find what is slowing the machine down
`kill`	Stops a process by its ID	To close a stuck program
`df -h`	Shows disk space in readable sizes	To check if a disk is nearly full
`du -sh`	Shows how big a folder is	To find what is using your space
`ip a`	Shows network interfaces and IP addresses	To find the IP of a machine
`ping`	Tests if another machine replies	To check that two machines can talk
`traceroute`	Shows the path packets take to a target	To see where a connection slows or stops
`nslookup`	Looks up a domain name's IP	To check that DNS is working
`dig`	Detailed DNS lookup	To see DNS records in more detail
`ss -tuln`	Lists open listening ports	To see which services are waiting for connections
`ssh`	Logs in to another machine securely	To control a VM or server remotely
`scp`	Copies files between machines over SSH	To send a file to or from a VM
`apt update`	Refreshes the list of available packages	Before installing new software on Ubuntu or Kali
`apt install`	Installs a package	To add a new tool

10. LinkedIn Posts

The brief asked for two LinkedIn posts. Both posts are published, and their links are in my README file.

Post	What it covers
Concept post	SOC analyst vs penetration tester: what is the difference? Explained in 5 points
Weekly post	A lab setup screenshot and my key learnings from the week, tagging DG Interns Hub

11. Challenges and How I Solved Them

Not everything worked on the first try. These are the problems I met, task by task, and what I did about each one.

Task	Challenge	How I solved it
Task 1: Lab	My laptop did not have enough storage for a Windows VM.	I used my Windows laptop as the third machine on the Host-only network and wrote this clearly in my documents.
Task 1: Lab	The login on the Kali VM failed with a Login incorrect message because I used the wrong username.	I tried again with the user kali and logged in.
Task 1: Lab	My first ping from Kali to the Windows laptop got no replies (4 sent, 0 received).	I ran the same ping again, and the second attempt got 4 replies with 0 percent packet loss.
Task 3: Network map	A traceroute command on Ubuntu gave the error Cannot handle -m option.	I ran traceroute -m 6 8.8.8.8 and it completed.
All tasks	Long text pasted into PowerShell got stuck.	I put long text into a file in Notepad instead of pasting it into the terminal.
Task 4: Bandit	In Level 0, I typed cat./- without a space, and it failed with No such file or directory.	I listed the folder with ls and used cat readme.
Task 4: Bandit	Levels 1 and 2 have files named - and --spaces in this filename--, which commands treat as options.	I used cat ./- for the first, and cat -- with quotes for the second.
Task 4: Bandit	In Level 3, my guesses .hidden, .Hidden and ./.hidden all failed.	I used ls -la, saw the real name ...Hiding-From-You and read it with cat.
Task 4: Bandit	My Bandit screenshots were on the Ubuntu side of my laptop.	I booted into Ubuntu and copied the screenshots into my Week 1 folder.

What these problems taught me

Most of my problems came from small typing mistakes or from commands reading a filename as an option. In each case the error message told me what was wrong, and I fixed it by changing one thing at a time. I also learned that it is better to say plainly when something differs from the brief, like the Windows VM, and to explain why.

12. Key Learnings

Before this week, Linux and networking felt like two separate subjects to me. Building the lab showed me how they connect. A ping from one machine to another travels through the same layers I studied in the OSI model, and a Host-only network showed me in practice how an isolated lab keeps vulnerable machines away from the real network.

Area	What I learned this week
Virtual lab	Creating VMs in VirtualBox, choosing a network mode and taking snapshots
Linux	Moving around the file system, searching with find and grep, managing users and groups, and setting permissions
Networking	Reading IP, gateway and DNS settings, using ping, traceroute, nslookup and dig, and the OSI and TCP/IP models

Area	What I learned this week
Security basics	The CIA triad, threat, vulnerability, risk and exploit, and the roles of SOC analyst, penetration tester and incident responder
Legal awareness	Authorization, scope, Rules of Engagement and the main cyber laws in India

13. Submission and Folder Structure

My work is in the Google Drive folder DG Interns Hub, inside the Week 1 folder, and I submitted the shared folder link through the submission form. The structure follows the brief:

Folder or file	Contents
Notes	8 theory notes, one file for each question
Writeups	TryHackMe writeups and Bandit level writeups
Findings, Lab Setup	VM list, IP addresses, pings and snapshots
Findings, Linux Practice	Users, groups, permissions, commands and the cron job
Findings, Network Map	ip, ping, traceroute, nslookup and dig screenshots and the draw.io diagram
Findings, Over the Wire Bandit	Screenshots of the Bandit levels
Cheat Sheet	Linux Command, What It Does, When to Use It
Weekly report, presentation and README	This report, my presentation and the README file

14. Conclusion

Week 1 gave me a strong start. I built a working lab, practised the Linux command line, mapped a real network, and solved the first ten Bandit levels. The theory notes and TryHackMe rooms gave me the background to understand what I was doing in the lab.

The hardest parts were the storage limit and a few small command mistakes. I now feel much more comfortable with Linux and networking. I want to build my career on the blue team side, especially SOC work, so I will use these foundations in the coming weeks.